Comparitive Performance Report for stretched-ec-16K-soko07-2+1-localize-15thOct vs stretched-ec-16K-soko07-2+1-balance-16thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

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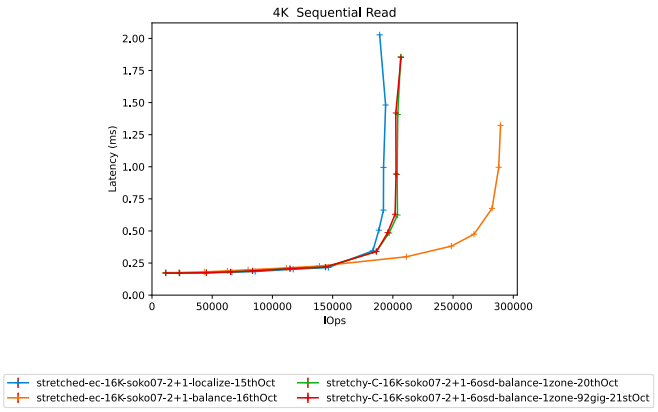
- Comparison summary for stretched-ec-16K-soko07-2+1-localize-15thOct vs stretched-ec-16K-soko07-2+1-balance-16thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct
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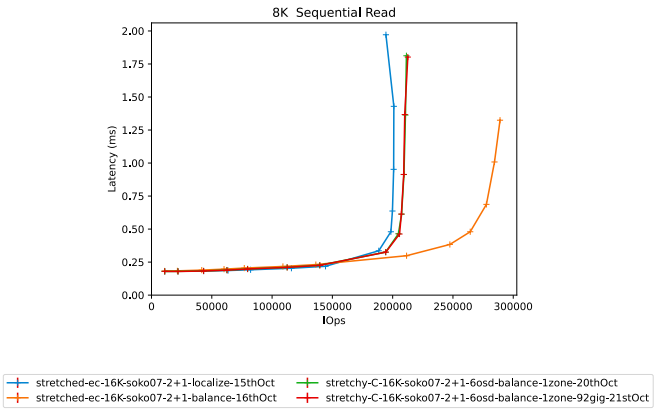
Comparison summary for stretched-ec-16K-soko07-2+1-localize-15thOct vs stretched-ec-16K-soko07-2+1-balance-16thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct vs stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

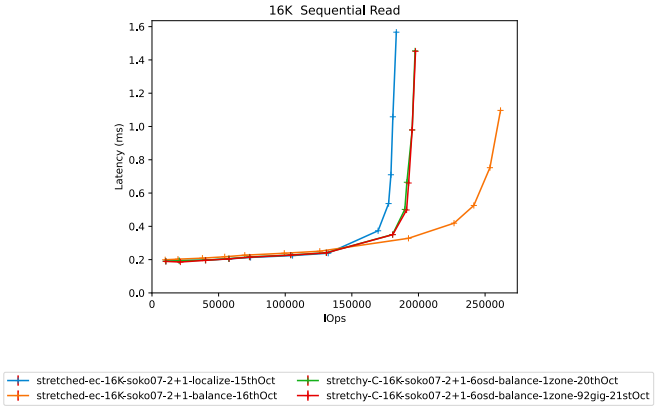
Response Curves

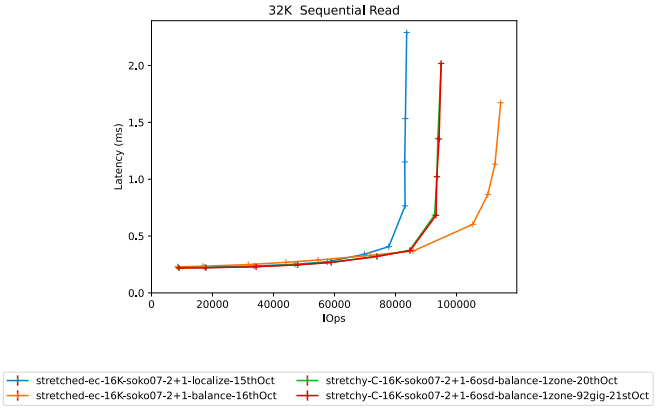
Number of Jobs 1

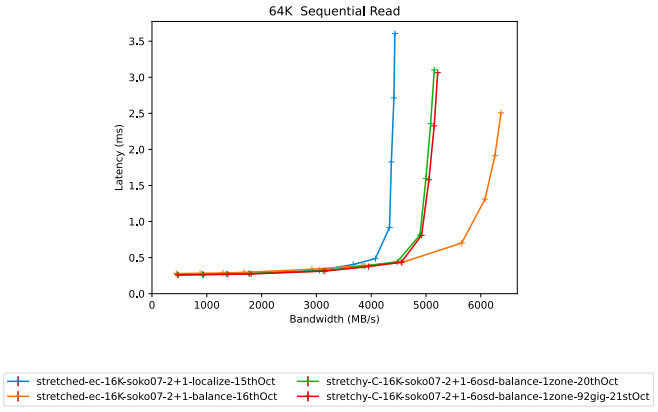
Sequential Read

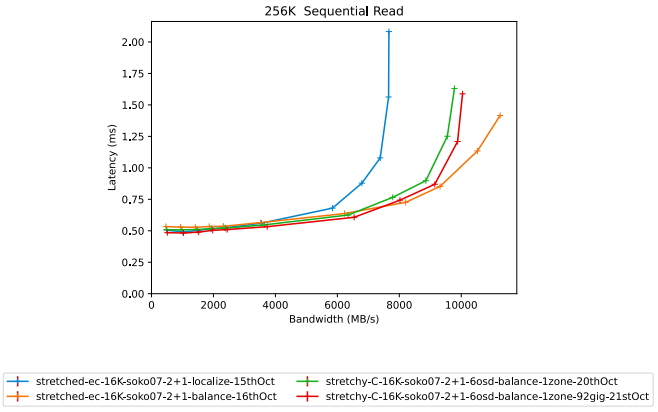


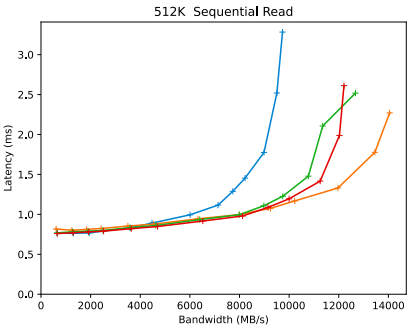


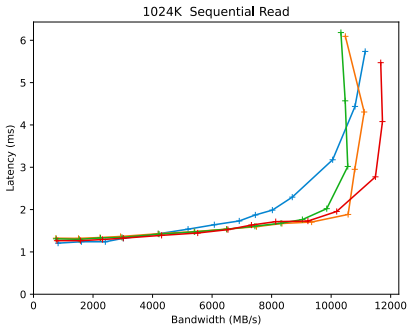






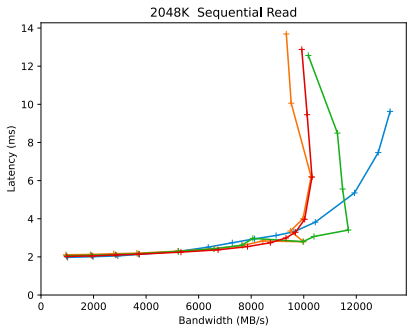


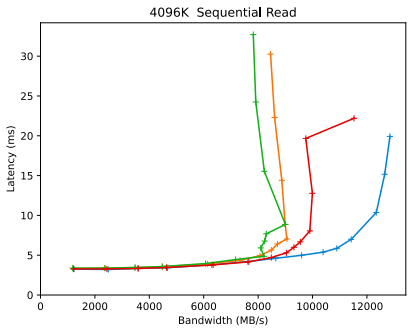








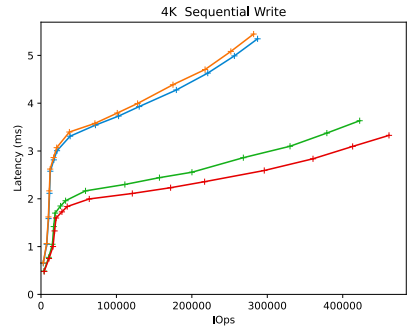


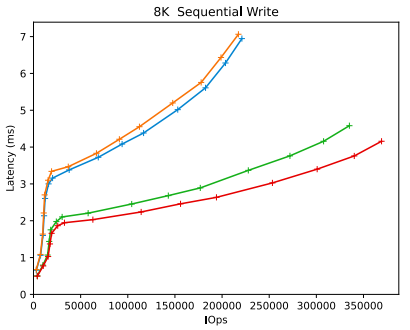






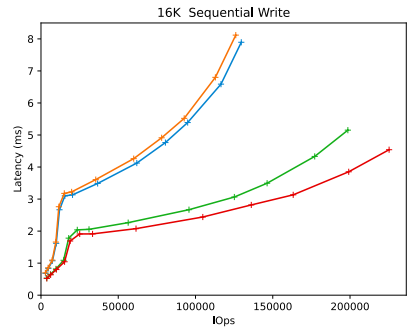
Sequential Write

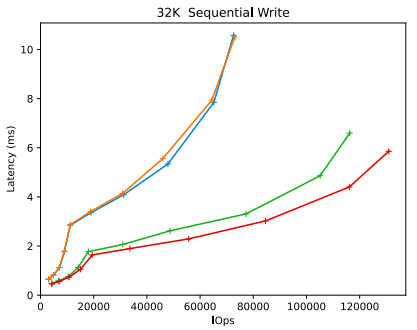






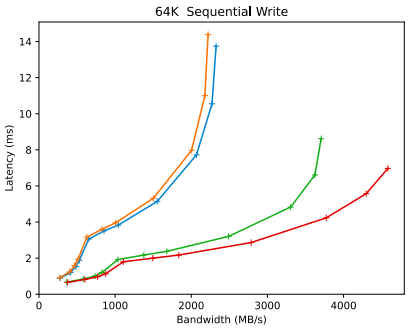


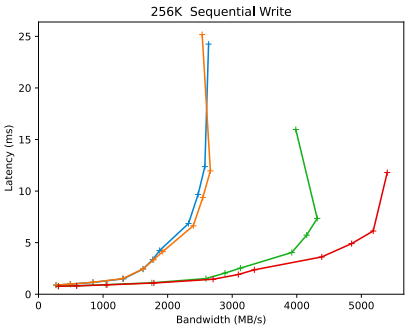






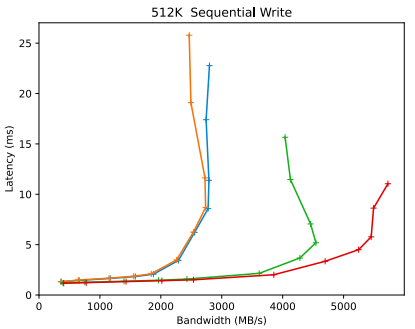


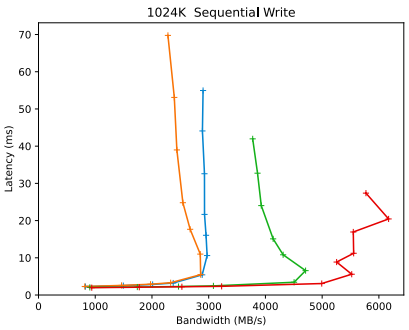

64K Sequential Write
Latency (ms)
Bandwidth (MB/s)


256K Sequential Write
Latency (ms)
Bandwidth (MB/s)


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

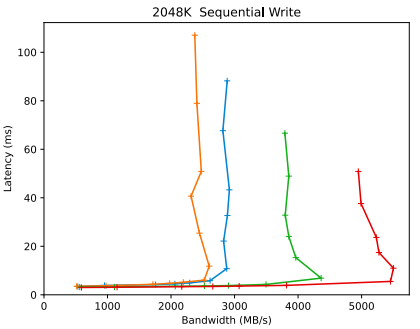

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

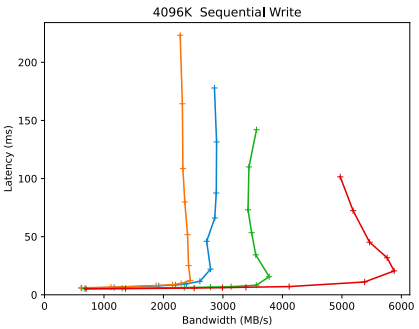

512K Sequential Write
Latency (ms)
Bandwidth (MB/s)


1024K Sequential Write
Latency (ms)
Bandwidth (MB/s)

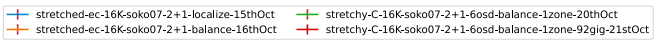

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

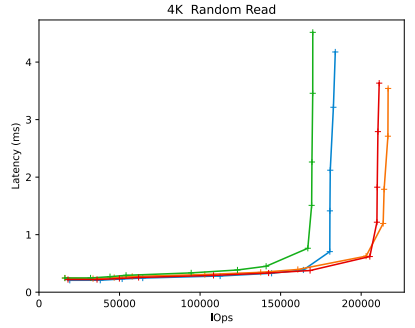

2048K Sequential Write
Latency (ms)
Bandwidth (MB/s)

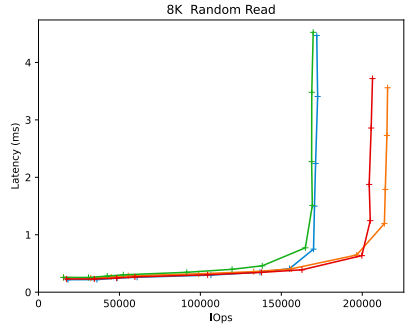

4096K Sequential Write
Latency (ms)
Bandwidth (MB/s)


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

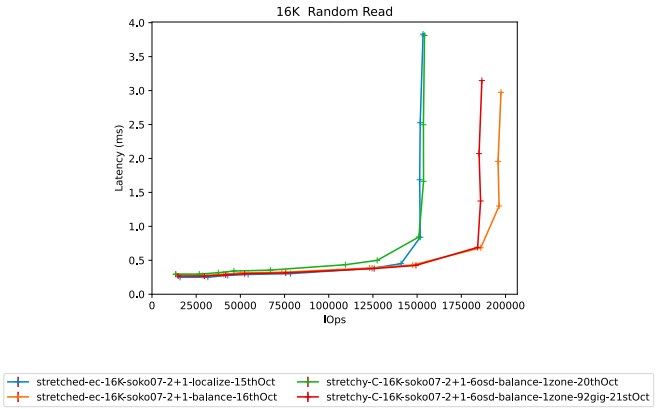
Random Read

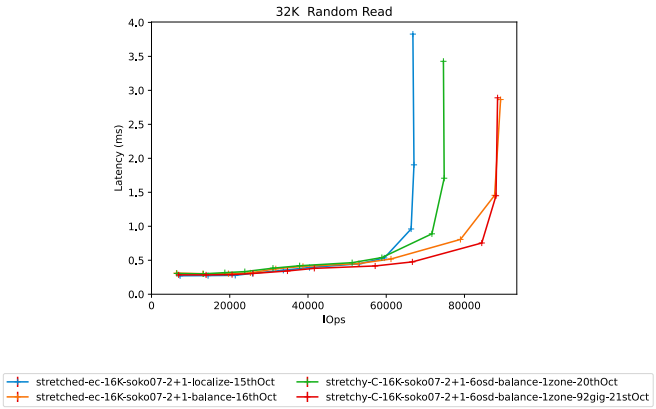

4K Random Read
Latency (ms)
IOps

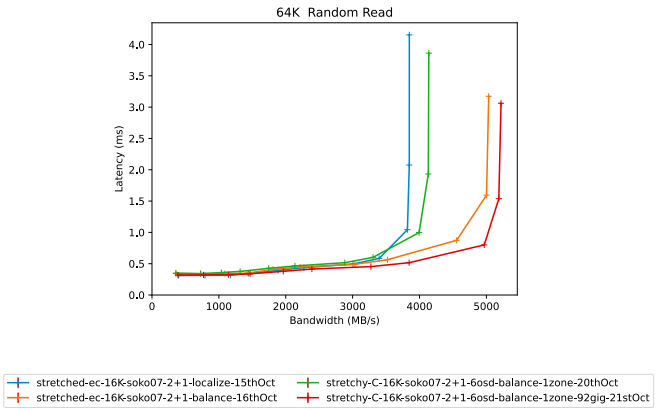

8K Random Read
Latency (ms)
IOps

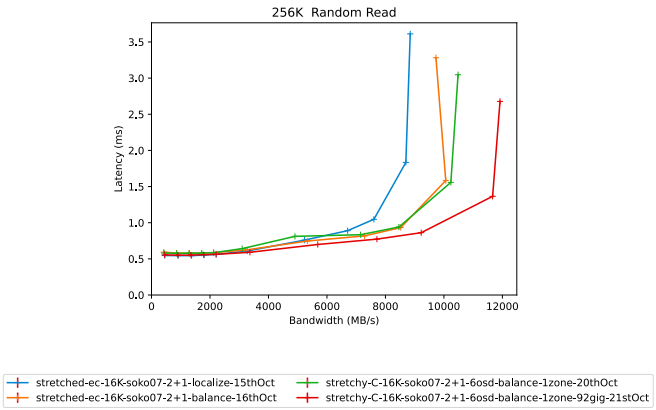

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

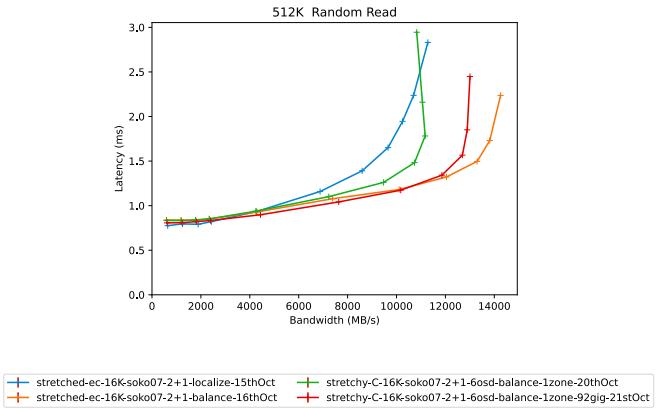

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

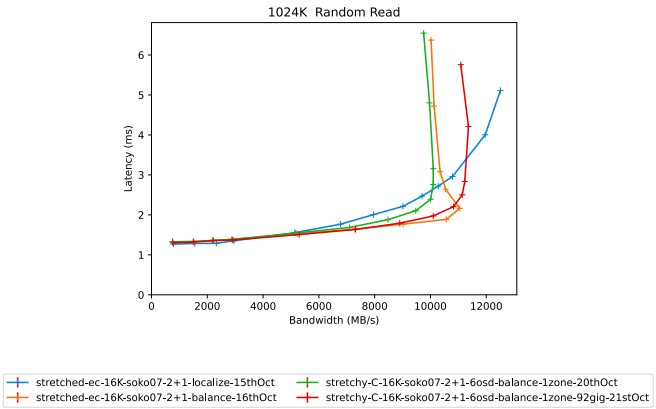


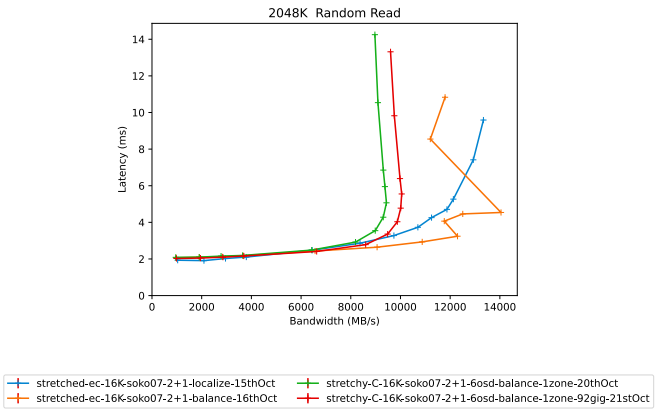


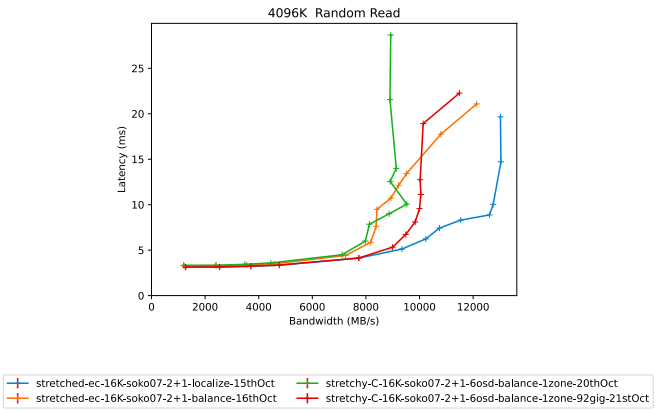




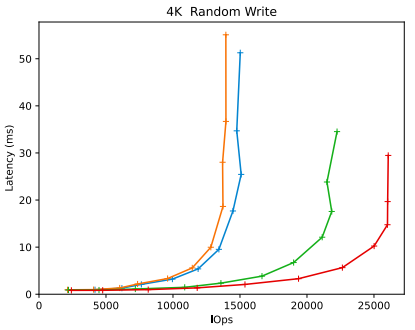


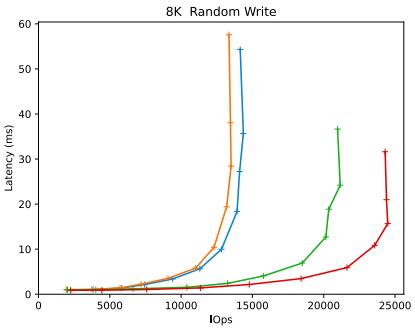






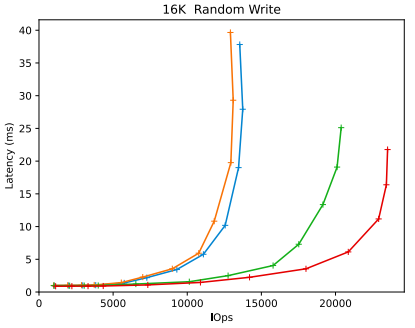
Random Write

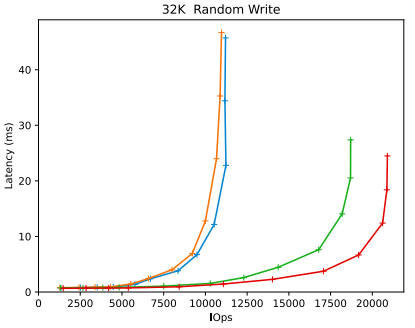

4K Random Write
Latency (ms)
IOps


8K Random Write
Latency (ms)
IOps


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

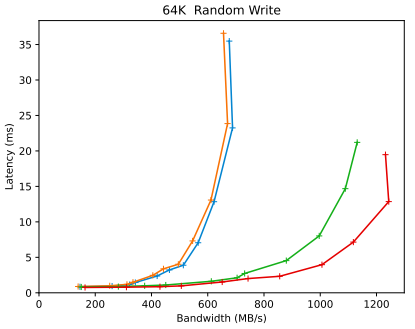

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

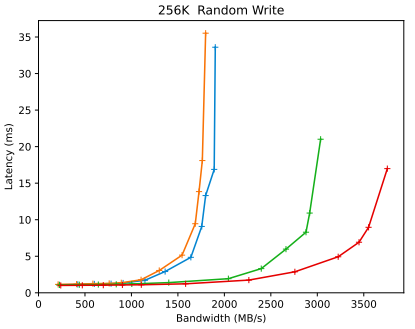

16K Random Write
Latency (ms)
IOps

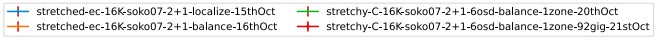

32K Random Write
Latency (ms)
IOps

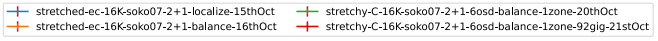

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

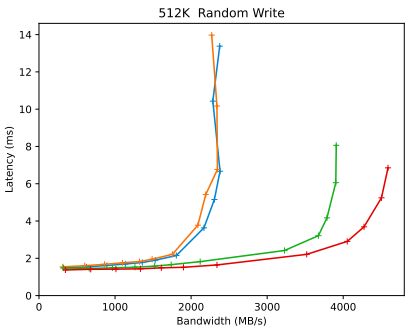

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

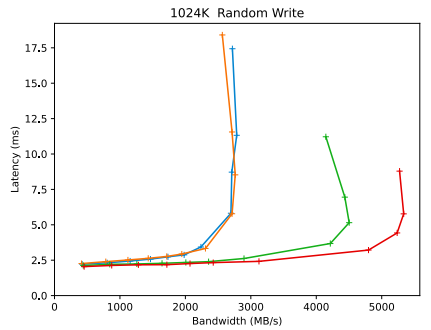

64K Random Write
Latency (ms)
Bandwidth (MB/s)

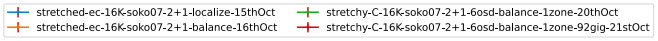

256K Random Write
Latency (ms)
Bandwidth (MB/s)

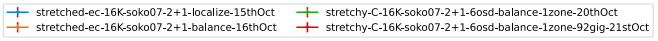

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

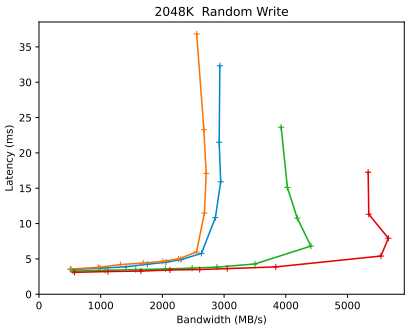

stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

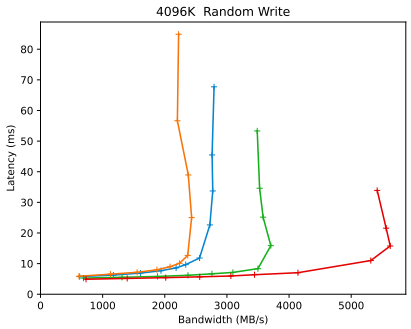

512K Random Write
Latency (ms)
Bandwidth (MB/s)


1024K Random Write
Latency (ms)
Bandwidth (MB/s)


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct


stretched-ec-16K-soko07-2+1-localize-15thOct
stretched-ec-16K-soko07-2+1-balance-16thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-20thOct
stretchy-C-16K-soko07-2+1-6osd-balance-1zone-92gig-21stOct

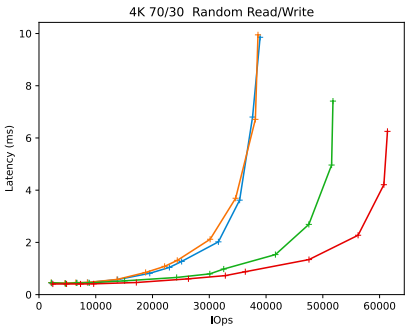


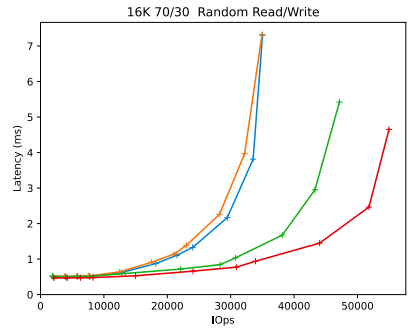




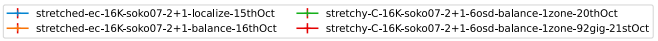


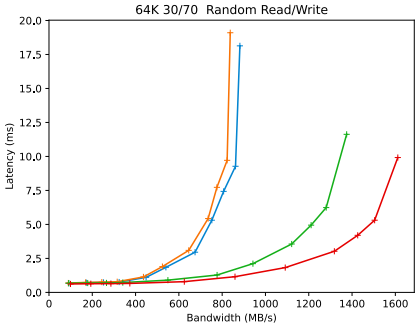
Random Read/Write

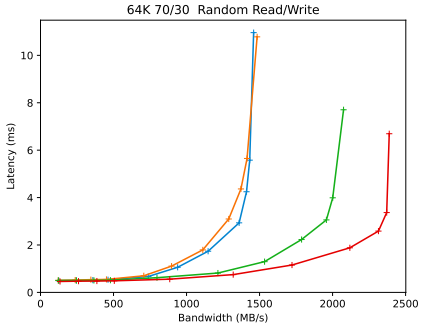
















Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
16krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 16384
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
1Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 1048576
```

```
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
1Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
  - 160
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
256krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 32
```

```
  - 64
  - 96
  - 128
  - 256
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
2Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
2Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
2Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
```

```
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 4
  - 5
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
32krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
```

```
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
4Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
4Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
4Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 4
  - 5
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
4k7030:
  jobname: randmix
```

```
    mode: randrw
    numjobs:
    - 1
    op_size: 4096
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  4krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  4krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
  512krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
```

```
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
512krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
  - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 12
  - 16
  - 20
  - 24
  - 32
512kseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
```

```
  - 96
  - 128
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
64krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
```

```
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 384
64kseqread:
  jobname: read
  mode: read
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 192
  - 256
64kseqwrite:
  jobname: write
  mode: write
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 8192
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
```

```
    - 128
    - 256
    - 384
    - 588
    - 768
  8krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq16kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
```

```
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
    - 1024
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
  seq256kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 256
  seq32kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
```

```
  seq32kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 512
    - 768
  seq4kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
    - 384
  seq4kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
    - 1024
    - 1280
    - 1536
  seq8kread:
    jobname: seqread
    mode: read
    numjobs:
```

```
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kwrite:
        jobname: seqwrite
        mode: write
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 8
        - 16
        - 24
        - 32
        - 48
        - 64
        - 128
        - 256
        - 384
        - 512
        - 768
        - 1024
        - 1280
        - 1536
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
```

```
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```

results

```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
2Mrandomread:
  jobname: randread
```

```
  mode: randread
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
2Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4194304
```

```
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
4Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 4194304
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
4k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4096
  total_iodepth:
  - 4
```

```
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 12
  - 16
  - 20
  - 24
  - 32
  - 48
  - 64
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
```

```
    - 32
    - 64
    - 96
    - 128
    - 256
  64k7030:
    jobname: randmix
    mode: randrw
    numjobs:
    - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 32
    - 64
    - 96
    - 128
    - 256
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 65536
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
  64kseqread:
    jobname: read
    mode: read
    numjobs:
    - 1
    op_size: 65536
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 192
    - 256
  8krandomread:
    jobname: randread
    mode: randread
```

```
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
```

```
      seq32kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 32768
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
      seq4kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 4096
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
```

```
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results


```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
2Mrandomread:
  jobname: randread
```

```
  mode: randread
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
2Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4194304
```

```
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  4Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
    - 32
    - 48
    - 64
  4k7030:
    jobname: randmix
    mode: randrw
    numjobs:
    - 1
    op_size: 4096
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  4krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 4
```

```
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 12
  - 16
  - 20
  - 24
  - 32
  - 48
  - 64
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
```

```
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
64kseqread:
  jobname: read
  mode: read
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 192
  - 256
8krandomread:
  jobname: randread
  mode: randread
```

```
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
```

```
      seq32kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 32768
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
      seq4kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 4096
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
```

```
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
2Mrandomread:
  jobname: randread
```

```
    mode: randread
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  2Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
    - 32
    - 48
    - 64
  32krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
  4Mrandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4194304
```

```
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  4Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
    - 32
    - 48
    - 64
  4k7030:
    jobname: randmix
    mode: randrw
    numjobs:
    - 1
    op_size: 4096
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  4krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 4
```

```
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 12
  - 16
  - 20
  - 24
  - 32
  - 48
  - 64
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
```

```
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
64kseqread:
  jobname: read
  mode: read
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 192
  - 256
8krandomread:
  jobname: randread
  mode: randread
```

```
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
```

```
      seq32kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 32768
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
      seq4kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 4096
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
```

```
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
2Mrandomread:
  jobname: randread
```

```
  mode: randread
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
2Mseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4194304
```

```
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  4Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
    - 32
    - 48
    - 64
  4k7030:
    jobname: randmix
    mode: randrw
    numjobs:
    - 1
    op_size: 4096
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  4krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 4
```

```
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
512kseqread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 12
  - 16
  - 20
  - 24
  - 32
  - 48
  - 64
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
```

```
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
64kseqread:
  jobname: read
  mode: read
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 192
  - 256
8krandomread:
  jobname: randread
  mode: randread
```

```
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
```

```
      seq32kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 32768
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
      seq4kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 4096
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
```

```
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```

results

```
  rawfio:
    block_devices:
    - /dev/nvme8n1
    - /dev/nvme8n2
    - /dev/nvme8n3
    - /dev/nvme8n4
    - /dev/nvme9n1
    - /dev/nvme9n2
    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
    - /dev/nvme14n4
    - /dev/nvme15n1
    - /dev/nvme15n2
    - /dev/nvme15n3
    - /dev/nvme15n4
    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
```

```
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  tmp_dir: /home/ljsanders/scripts/nvme/tmp
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
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results

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    block_devices:
    - /dev/nvme8n1
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    - /dev/nvme9n3
    - /dev/nvme9n4
    - /dev/nvme10n1
    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
    - /dev/nvme11n1
    - /dev/nvme11n2
    - /dev/nvme11n3
    - /dev/nvme11n4
    - /dev/nvme12n1
    - /dev/nvme12n2
    - /dev/nvme12n3
    - /dev/nvme12n4
    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
    - /dev/nvme14n1
    - /dev/nvme14n2
    - /dev/nvme14n3
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    direct: 1
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    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
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  osds:
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  user: root
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  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
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    - /dev/nvme10n2
    - /dev/nvme10n3
    - /dev/nvme10n4
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    - /dev/nvme12n4
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    - /dev/nvme13n2
    - /dev/nvme13n3
    - /dev/nvme13n4
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    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randread
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 100
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
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  - --- server1 ---
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    - /dev/nvme13n1
    - /dev/nvme13n2
    - /dev/nvme13n3
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    client_ra: 128
    concurrent_procs: 32
    direct: 1
    fio_cmd: /usr/local/bin/fio
    iodepth:
    - 32
    - 48
    ioengine: libaio
    iterations: 1
    mode:
    - randwrite
    numjobs: 1
    op_size:
    - 4096
    osd_ra:
    - 4096
    ramp: 30
    time: 90
    vol_size: 50000
cluster:
  archive_dir: /home/ljsanders/scripts/nvme/tmp
  clients:
  - --- server1 ---
  head: --- server1 ---
  iterations: 1
  osds:
  - --- server1 ---
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    - /dev/nvme13n3
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